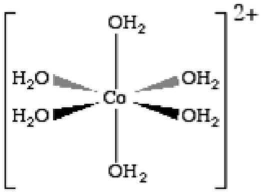
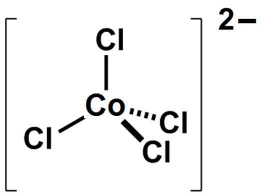
Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks available					
9.	(a)			 (1)  (1)	2			2		
	(b)			ligands cause d-orbitals to split into three lower and two higher energy levels (1) electrons absorb energy and are promoted to a higher energy level (1) colour seen is the colours not absorbed (1)	3			3		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)			white precipitate (1)		1				1
				silver chloride forms / silver ions cause a precipitate with chloride (1)		1				1
				solution turns pink (1)			1			
				removal of chloride causes equilibrium to shift to left (1)			1	4		
	(d)	(i)		$f = 5.79 \times 10^{14}$ (1)						
				$E = hf = 3.84 \times 10^{-19}$ (1)						
				$E = 231$ (1)		3		3	3	
		(ii)		value above 600 nm as this is absorbed by chloro but not aqua / value below 560 nm as this is absorbed by aqua complex but not chloro		1		1		1

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		$K_c = \dfrac{[[CoCl_4]^{2-}] \times [H_2O]^6}{[[Co(H_2O)_6]^{2+}] \times [Cl^-]^4}$ (1) $mol^2\ dm^{-6}$ (1)		2		2	1	
		(iv)		concentration of aqua complex = 0.596 concentration of chloride = 0.224 / concentration of chloro complex = 0.124 award (2) for all three concentrations correct award (1) for any two correct K_c = 14.0 (1)			3	3	3	
				Question 9 total	**5**	**8**	**5**	**18**	**7**	**3**